

NI 43-101 Technical Report

on the

Patterson Lake South Property

Athabasca Basin, Saskatchewan, Canada

Prepared for: Fission Uranium Corp.

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Qualified Persons:

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1. Summary

Patterson Lake South (PLS) is an advanced-stage uranium exploration project located in the Athabasca Basin of northern Saskatchewan, Canada. The property is host to the Triple R deposit, which contains significant high-grade uranium mineralization at the unconformity between Athabasca Group sandstones and Paleoproterozoic basement rocks. Current mineral resource estimates include Indicated resources of 5.2 million tonnes at 1.52% U_3O_8 , containing approximately 174 million pounds of uranium. Inferred resources are 3.8 million tonnes at 0.85% U_3O_8 , containing approximately 71 million pounds of uranium. Mineralization is associated with structurally controlled alteration zones in basement rocks, with observed grades exceeding 50,000 ppm U_3O_8 in select intervals. The deposit shows characteristics comparable to other world-class unconformity-type uranium systems in the Athabasca Basin.

2. Introduction

This technical report has been prepared in accordance with the Canadian Securities Administrators' National Instrument 43-101 Standards of Disclosure for Mineral Projects (NI 43-101) and Form 43-101F1. The report summarizes exploration activities, drilling results, and mineral resource estimates for the Patterson Lake South property. A site visit to the PLS property was conducted during May 2024 by both Qualified Persons. Data sources include 47 diamond and reverse-circulation drill holes completed between 2013 and 2024, historical and recent airborne geophysical surveys, surface geochemical programs, and detailed geological mapping. All historical data was reviewed and verified against original source documents.

3. Reliance on Other Experts

This technical report has been prepared by the Qualified Persons listed herein. Information regarding the mineral tenure, claims ownership, and regulatory status of the property was provided by Fission Uranium Corp. legal counsel and has not been independently verified by the Qualified Persons. The Qualified Persons have relied upon these representations and do not assume responsibility for errors or omissions in tenure information. All other technical and geological interpretations are based on the Qualified Persons' direct review of assay data, drill logs, and core photography.

4. Property Description and Location

The Patterson Lake South property comprises 31,039 hectares of contiguous mineral claims distributed across 17 staked claims in the Athabasca Basin of northern Saskatchewan. The property is centered at approximately UTM Zone 13N coordinates 495,000 easting and 6,220,000 northing (NAD83). Access to the project is achieved via gravel road from the community of Points North Landing, approximately 45 kilometres to the southwest. The property is situated in a subarctic continental climate zone at elevations ranging from 420 to 450 metres above sea level. The claims are held in good standing with no known encumbrances, liabilities, or third-party royalties, except for a standard 3% net smelter return (NSR) royalty retained by the Crown. The property is accessible by road during the winter months via ice road to Stony Rapids, the nearest year-round settlement, located approximately 140 kilometres to the north.

5. Accessibility, Climate, Local Resources, Infrastructure and Physiography

The Patterson Lake South property is situated in a remote wilderness area of northern Saskatchewan. The region is characterized by subarctic continental climate with long, cold winters and short summers. The field season is restricted to June through September due to seasonal ice road access and permafrost constraints. The property is covered by mature boreal forest dominated by black spruce and jack pine with thick moss and lichen ground cover. Surface topography is relatively subdued with numerous lakes and muskeg. The nearest permanent community is Stony Rapids, accessible year-round by air charter, with winter road access via seasonal ice roads. A gravel airstrip capable of handling Twin Otter and larger aircraft is located at Points North Landing, 45 kilometres from the property. Electricity is generated by diesel at the property, and potable water is obtained from local lakes. The region has limited infrastructure, necessitating self-sufficiency in project operations including accommodation, food supply, and emergency medical response.

6. History

Initial exploration for uranium on the Patterson Lake South property was conducted by SMDC (Saskatchewan Mineral Development Company) during the 1970s following regional airborne geophysical surveys. Limited exploration activity occurred until 2013 when Fission Uranium Corp. discovered significant uranium mineralization in drill hole PLS13-022 at the unconformity. This discovery prompted a systematic delineation program conducted between 2013 and 2018, resulting in the definition of the Triple R deposit. Major drilling and resource definition activities were conducted from 2014 through 2018. The current exploration program, initiated in 2018, has focused on upgrading the mineral resource classification from inferred to indicated through infill drilling and detailed geological characterization. A total of 47 drill holes have been completed to date on the property, with cumulative drilling exceeding 15,000 metres.

7. Geological Setting and Mineralization

The Patterson Lake South property is host to unconformity-type uranium mineralization characteristic of the Athabasca Basin. Basement lithologies are predominantly Paleoproterozoic pelitic and semipelitic gneisses intruded by graphitic metapelite units. These basement rocks are unconformably overlain by Athabasca Group sandstones at depths ranging from 50 to 150 metres. Mineralization is associated with structurally controlled alteration zones developed in the basement rocks adjacent to and below the unconformity surface. Key geological controls on mineralization include: (1) proximity to conductive graphitic basement units, (2) development of hematite and illite alteration halos, (3) intersection of brittle basement faults with the unconformity surface, and (4) presence of clay-rich sandstone units directly above the unconformity. High-grade mineralization ($>5\% \text{ U}_3\text{O}_8$) has been observed in multiple drill holes, with peak grades exceeding 52,000 ppm U_3O_8 recorded in hole PLS-22-08 from 398 to 401 metres depth. Mineralization comprises uraninite and coffinite minerals intergrown with pyrite and associated with clay alteration.

8. Deposit Types

The Triple R deposit is classified as a classic unconformity-related uranium deposit, the same deposit type as several of the world's largest and highest-grade uranium systems. Comparable deposits in the Athabasca Basin include McArthur River (Cameco, 600+ million pounds U_3O_8 reserves), Cigar Lake (Cameco), and the nearby Arrow deposit (NexGen Resources). Unconformity-type uranium deposits are formed by precipitation of uranium from migrating groundwaters where chemical reducing agents (commonly organic carbon and pyrite) in basement rocks create a geochemical trap. Key geological controls on the PLS deposit include: structural dilation zones at the unconformity interface, graphitic conductors in the basement that serve as fluid conduits, and hydrothermal alteration creating permeable pathways for ore fluids. The deposit has not yet undergone feasibility studies, and the current resource is classified as both Indicated (67% of tonnes) and Inferred (33% of tonnes) in accordance with CIM 2014 Mineral Resource and Reserve Definitions.

9. Exploration

Exploration methodologies employed on the Patterson Lake South property have included: (1) airborne VTEM (Very Low Frequency electromagnetic) geophysics flown at 25-metre line spacing to define conductive basement targets, (2) ground-based time-domain electromagnetic (TDEM) surveys over selected high-priority zones, (3) soil and lake sediment geochemistry sampling to define pathfinder element anomalies, (4) detailed geological mapping and prospecting, and (5) diamond and reverse-circulation drilling. The VTEM survey identified multiple basement conductors, several of which have been drill-tested. Airborne radiometric data shows uranium and potassium enrichment above the unconformity in areas of known mineralization. Soil geochemistry displays multi-element anomalies including elevated molybdenum, vanadium, and selenium in association with known uranium prospects. To date, 47 drill holes totalling approximately 15,000 metres have been completed, with 10 holes (3,645 metres) completed in the current 2018-2024 program. All drilling has been conducted to core refusal or planned termination depth.

10. Drilling

The current exploration program (2018-2024) comprises 10 diamond and reverse-circulation drill holes totalling 3,645 metres. All drilling was conducted by an independent, industry-standard drilling contractor using truck-mounted or skid-mounted rigs. Diamond drilling employed core sizes ranging from HQ to NQ depending on hole conditions and planned final depth. All drill holes were collared with measured UTM coordinates and elevations using differential GPS. Core was logged in detail by project geologists at the drill site, photographed both dry and wet, and stored in secure core logging facilities. Geotechnical parameters including recovery and RQD (rock quality designation) were recorded for all core intervals. Key mineralized intercepts include: (1) hole PLS-22-08 with 52,000 ppm U_3O_8 from 398-401 metres, (2) hole PLS-21-05 with 8,500 ppm U_3O_8 over 12 metres, and (3) hole PLS-23-04 with multiple lower-grade intervals $>0.5\%$ U_3O_8 . Average hole depth in the current program is 364.5 metres.

11. Sample Preparation, Analyses and Security

Core samples selected for uranium assay were cut in half using a motorized core saw. Sample intervals ranged from 0.5 to 2.0 metres depending on lithology and visible mineralization. One half of the core was retained for archival purposes; the other half was placed in pre-numbered assay bags. All samples were shipped under chain-of-custody documentation to SRC Geoanalytical Laboratories in Saskatoon, Saskatchewan, a commercial accredited laboratory. Uranium was assayed by inductively coupled plasma mass spectrometry (ICP-MS) following partial acid digestion. Detection limit is 10 ppm U_3O_8 . Quality assurance and quality control (QA/QC) procedures included: (1) one blank sample per 20 submitted samples, (2) one duplicate sample per 20 submitted samples, (3) insertion of standard reference materials with known uranium concentrations. Statistical analysis of QA/QC results indicates no material issues with sample contamination, cross-contamination, or analytical accuracy. All assay data has been entered into a secure, password-protected relational database with version control and audit trails.

12. Data Verification

Both Qualified Persons conducted an on-site inspection of the Patterson Lake South property during May 2024. During this visit, core from selected drill holes was examined in detail, including visual inspection of high-grade intervals. Drill core from holes PLS-22-08, PLS-21-05, and PLS-23-04 was re-sampled in selected intervals to verify previous assay results. An independent review of the complete historical assay database was conducted, with cross-checking of approximately 50 individual drill core assay intervals against the original laboratory assay certificates on file. No discrepancies of material significance were identified between the assay database and the source laboratory reports. Collar locations for all drill holes were independently verified using differential GPS. Historical geological logs from the 2013-2018 program were reviewed and compared against core photographs and the current geological interpretation. The Qualified Persons are satisfied that the data underlying the resource estimation is reliable and of sufficient quality to support a technical report in accordance with NI 43-101.

13. Mineral Resource Estimates

Mineral resource estimates were prepared by Qualified Person Dr. Sarah Thompson using industry-standard ordinary kriging methodology. The resource estimation was conducted using a parent cell size of 10 metres by 10 metres by 3 metres (vertical). A minimum cut-off grade of 0.2% U_3O_8 was applied to the inferred resource and 0.5% U_3O_8 to the indicated resource, reflecting anticipated mining and milling parameters. The Indicated resource is estimated at 5.2 million tonnes at an average grade of 1.52% U_3O_8 , containing approximately 174 million pounds of contained uranium. The Inferred resource is estimated at 3.8 million tonnes at an average grade of 0.85% U_3O_8 , containing approximately 71 million pounds of contained uranium. Resource estimates are reported in accordance with the Canadian Institute of Mining, Metallurgy and Petroleum (CIM) Definition Standards for Mineral Resources and Reserves, as adopted by the CIM in 2014. The Indicated resource is confined to areas within 50 metres of drill holes with measured assay intervals; the Inferred resource extends to 150 metres from drill hole control. No Measured resources have been declared.

14. Adjacent Properties

The Patterson Lake South property is located in a region with multiple other uranium exploration projects at various stages of development. The Denison Mines Ltd. Wheeler River project is located approximately 40 kilometres to the east and hosts the Phoenix deposit, also an unconformity-type uranium system. NexGen Resources' Arrow deposit is located approximately 8 kilometres to the south and is one of the largest high-grade uranium deposits ever discovered in the Athabasca Basin with announced resources exceeding 940 million pounds of U_3O_8 . The Cameco-operated McArthur River mine is approximately 65 kilometres to the northeast. These adjacent properties demonstrate the regional prospectivity of the area for unconformity-type uranium mineralization. The Patterson Lake South property benefits from this regional infrastructure and exploration support, which has facilitated data access and geological understanding of the local geological framework.

15. Other Relevant Data and Information

Environmental baseline studies are currently being conducted on the Patterson Lake South property under a federal environmental assessment protocol. To date, baseline water quality, soil, and fish tissue samples have been collected and analyzed. No material environmental concerns have been identified at this early stage. Indigenous consultation and accommodation activities with local First Nations have been initiated as of 2023, as required by Crown consultation policy. A preliminary archaeology assessment has been completed with no documented archaeological or heritage sites recorded on the property. The property is located within the traditional territories of the Fond du Lac Dene Nation and Athabasca Chipewyan First Nation. Ongoing consultation with these nations is planned to address any concerns and to seek community support for continued exploration. Climate change considerations specific to the subarctic region, including permafrost stability and access timing, have been factored into the exploration and development planning. No material issues have been identified that would restrict continued exploration activities on the property.

16. Interpretation and Conclusions

The Patterson Lake South property is host to a significant high-grade uranium resource within the Triple R deposit. The deposit displays the geological characteristics of a classic unconformity-related uranium system, comparable to the world-class deposits of the Athabasca Basin. The defined Indicated and Inferred resources represent approximately 245 million pounds of contained uranium, positioning PLS as a material uranium asset for Fission Uranium Corp. The current resource classification is based on 47 drill holes distributed over the deposit area, with Indicated resources defined within areas of good drill control and Inferred resources extending to edge-of-envelope areas. The Qualified Persons believe that the geological setting, mineralization characteristics, and historical exploration data support the interpretation that significant additional mineralization exists beyond the current resource boundary. Continued systematic exploration, including deep drilling in areas of basement conductor anomalies, is warranted to test for additional high-grade zones and to expand the resource footprint. The current deposit geometry and grade distribution suggest that further infill drilling will likely upgrade a portion of the Inferred resource to Indicated classification.

17. Recommendations

The Qualified Persons recommend a two-phase exploration program designed to upgrade resource classification and expand the resource footprint. Phase 1 (Calendar Year 2025) should comprise approximately 12,000 metres of infill diamond drilling targeting areas of Inferred resource adjacent to current Indicated resource blocks, with the objective of upgrading a substantial portion of inferred tonnes to Indicated classification. Estimated budget for Phase 1: CAD \$8.5 million. Phase 2 (Calendar Year 2026) should comprise approximately 8,000 metres of step-out drilling to test deep basement conductor anomalies and to extend mineralization beyond the current resource boundary, with emphasis on areas showing early-stage geochemical and geophysical indicators of mineralization. Estimated budget for Phase 2: CAD \$6.2 million. Both phases should include ongoing environmental baseline studies, detailed mineralogical characterization, and preliminary metallurgical test work to support potential future feasibility studies. The combined Phase 1 and Phase 2 program investment of CAD \$14.7 million is expected to provide sufficient data quality and resource definition to support scoping-level mining and economic studies by the end of 2026.